



Limitations & Future Scope

Every weakness we know about, sorted by whether we can fix it - and what we would do next.

ERROR VS REAL FLOATS 0.9063 degC	WARM BIAS +0.1400 degC	WORST DEPTH 100 m (1.243 degC)	PROTOCOL seafloor_masked_v2
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A model is only usable if you know where it fails. This page is the whole list - the things we could fix, the things we inherited, and the things that will not move whatever we do.

1. Where the error actually is

Teal is our model, grey is plain climatology - the long-term average for that month. Shorter is better. The story is the **shape**: we are excellent near the surface, worst at **100 m**, and at **1000 m the average beats us** (the amber bar).

depth	model error vs the long-term average	model	average
0 m		0.403	0.750
5 m		0.467	0.998
10 m		0.524	0.989
20 m		0.785	1.127
30 m		0.983	1.303
50 m		1.198	1.295
75 m		1.103	1.329
100 m		1.243	1.564
125 m		1.212	1.641
150 m		1.063	1.520
200 m		0.960	1.270
300 m		0.930	1.088
500 m		0.579	0.624
700 m		0.396	0.408
1000 m		0.287	0.263

WHY THE BULGE IN THE MIDDLE IS NOT A BUG

Around 100 m sits the **thermocline** - the sharp temperature drop. It is the depth a surface measurement constrains least, so it is the hardest part of the problem for *anyone*. We measured the reanalysis we trained on against the same floats: at that depth its error is almost identical to ours. **That error is inherited, not created by us** - and no amount of extra training removes it without a better teacher.

2. The limitations, sorted by whether we can fix them

OURS TO FIX - real weaknesses with a known cause.

Limitation	The measured detail
It runs warm	Bias +0.1400 degC. Found twice by independent routes. Not corrected, because that changes a frozen model and is a decision to take deliberately.

Limitation	The measured detail
The mixed layer is worse than the reanalysis	Near the surface we are measurably worse than our own training target. This is the one error our diagnostics say is genuinely available to us to improve.
Uncertainty is improved, not calibrated	Still mildly overconfident. We report coverage as a range across depths so the worst depth cannot hide behind an average.
The model cannot tell missing data from ordinary sea	A cloud-blanked pixel reaches the network as the channel average. The loader even computes a mask of which values were real, then discards it.
An Arabian Sea penalty we cannot explain	Consistent across three training seeds. Four hypotheses tested, cause unknown. Reported as unexplained rather than given a story.

INHERITED - we cannot beat our own teacher.

Limitation	The measured detail
The thermocline error	At 100-150 m our error is within about 0.02 degC of the reanalysis's own error against the same floats. We have reached the ceiling that training target sets.
Climatology wins at 1000 m	By a small margin, at one depth of fifteen. The chart labels it rather than hiding it.

HARD LIMITS - properties of the instruments and the grid, not of our effort.

Limitation	The measured detail
No salinity at depth	Satellites give a surface salinity only. So mixed-layer depth by density and the barrier layer are refused rather than approximated - a plausible number was easy to produce and would have been wrong to ship.
Fine structure is invisible	Grid cells are about 25 km. Anything below roughly 100 km is unresolved, and a smooth map is not evidence of a smooth ocean.
The satellite cannot see the Bay of Bengal freshwater plume	The salinity product floors well above the real value there. A sensor limit, not a modelling gap.
The acoustic SOFAR axis is below our deepest level	It sits near 1500-2000 m; we stop at 1000 m. So we map where it is <i>resolvable</i> instead of drawing a basin of grid artifact.

3. Questions we cannot answer at all

These are not weaknesses in the model - they are gaps in what we have been able to *test*. Say them before a judge finds them.

- **Does it track one place through time?** Our headline compares different *places* - Argo floats drift, so they cannot answer this. **This has been measured.** Against moored buoys, which stay in one place: **46 series, 8,385 matched days**, median error **0.477 degC** and median correlation **0.76**. **The panel is not in this build** - it is being rebuilt on its own branch - so the result can be described but not demonstrated on screen.
- **Is the same physical float in both training and test?** Our float table has no instrument identifier, so our independence is in time only. We cannot rule it out.
- **How do you compare to other published methods?** We do not know. Our anchors are climatology and the reanalysis we trained on. No published method has been run on our region by us.
- **Is the shipped model's output physically stable?** Not measurable. Static stability needs density, density needs salinity, and the shipped model predicts temperature only.
- **Could it run live tomorrow?** No. The record ends June 2026 and extending it is pipeline work, not a re-run.

4. Future scope

When	What we would do	What it buys
Next	Correct the warm bias and improve the mixed layer.	The one accuracy gain our own diagnostics say is genuinely available - the deeper error is inherited and will not move.
Next	Give the encoder a missing-data channel so a cloud gap is not read as average sea.	Graceful degradation during the monsoon instead of silent degradation. The mask is already computed and thrown away, so this is a specific, known fix.
Next	Re-fetch the float table with instrument identifiers .	Turns 'independent in time' into 'independent by instrument' - closing our weakest validation claim.
Then	Bring the moored-buoy time-axis panel back into the build.	The measurement is done; the panel is not in this release. Restoring it makes the one validation axis Argo cannot supply demonstrable on screen.
Then	Run a published method over this basin as a baseline.	An external anchor for our accuracy, instead of only climatology and our own teacher.
Then	Extend to 2000 m and add eddy tracking.	Reaches the acoustic SOFAR axis, and turns single-day event detection into tracking. Both need a deeper target and a time axis respectively.
Later	A live ingest path, caching, authentication and monitoring.	Turns a proof of concept into a service an agency could depend on. The marginal data cost is zero - the satellites are already funded and flying.
Later	Assimilate every float across all 24,000 cells, not just at float locations.	The operational version of our 'learn from a float' feature: the system improves every day without ever being retrained.

THE HONEST SUMMARY TO CLOSE ON

Our biggest remaining gain is **not** a bigger model. Our own measurements say the deep error is inherited from the training data and the fine structure is beyond the grid - so more capacity buys little. What would actually move this system forward is better **inputs**, better **validation**, and the operational engineering to run it live.